

7.1) 4) $a, b \in \mathbb{Z}^+$, $L = \{n \in \mathbb{Z}^+ \mid \exists s \in \mathbb{Z} \exists t \in \mathbb{Z} (n = sa + tb)\}$ — (1)

By well Ordering principle, L has a smallest element.

Let that be d so $\forall x \in L (d \leq x)$ — (2)

$$\longrightarrow d = \gcd(a, b)$$

We will show that $d|a \wedge d|b \wedge \forall c \in \mathbb{Z}^+ [(c|a \wedge c|b) \rightarrow c \leq d]$

Let q_1 & r_1 be quotient & remainder when a is divided by d .

$$\text{so } a = q_1 d + r_1 \wedge q_1, r_1 \in \mathbb{N} \wedge 0 \leq r_1 < d. \text{ — (3)}$$

We prove by Contradiction. Supp $r_1 > 0$ — (4)

Note since $d \in L$, $\exists s \exists t \in \mathbb{Z} (d = sa + tb)$ — (5)

$$\text{From (3)} \rightarrow r_1 = a - q_1 d = a - q_1 (sa + tb) \quad \therefore \text{ by (5)}$$

$$= a - q_1 sa - q_1 tb$$

$$= (1 - q_1 s)a - (q_1 t)b$$

Since $q_1 \in \mathbb{N}$ & $s, t \in \mathbb{Z}$ so $1 - q_1 s \in \mathbb{Z}$ & $q_1 t \in \mathbb{Z}$. So r_1 can be written as linear combination of a & b .

So $r_1 \in L$. But we know $r_1 < d$. This contradicts (2), that d was smallest element of L . Therefore it cannot be case that $r_1 > 0$, so $r_1 = 0$. So $a = q_1 d + 0 \rightarrow a = q_1 d$ so $d|a$. Similar reasoning shows $d|b$.

Now let $c \in \mathbb{Z}^+$ be arbitrary. Supp $c|a$ & $c|b$. So c would divide any linear combination of a & b (Just Next Theorem proves this). so $c|(sa + tb) = d$. Since $c|d$ so $c \leq d$.

So d is greatest element such that $d|a$ & $d|b$ so

$$d = \gcd(a, b).$$

$$d|a \wedge d|b \quad \longrightarrow \quad \forall s \in \mathbb{Z} \forall t \in \mathbb{Z} [d|sa + tb]$$

$$\exists k \in \mathbb{Z} (dk = a) \text{ — (1)}$$

$$\exists l \in \mathbb{Z} (dl = b) \text{ — (2)}$$

$$\textcircled{1} \rightarrow dks = as \quad \& \quad \textcircled{2} \rightarrow dlt = bt$$

$$\text{so } as + bt = dks + dlt = d(ks + lt) \quad \therefore \text{so } d | (as + bt)$$

This proves in general, if $a|b$ & $a|c$ then a divides any linear combination of b & c .

The Reverse is not true. That is,

$$\text{if } d | (as + bt) \not\rightarrow d|a \& d|b$$

To see this, (supp) let $a = 4$, $b = 6$, let $d = 2$.

$$2 \times 2 + 3 \times 2 = 4 + 6 = 10 \text{ is a linear combination of } a \& b \& d | 10 \text{ but } d \nmid b = 6$$

so if a divides a linear combination of b & c , then it need not be that $a|b$ & $a|c$.

But this theorem holds if $d = \gcd(a, b)$. Then d itself is linear combination of a & b . Also if any $c|d$ then $c|a$ & $c|b$.

$$\text{Proof: } \text{supp } c | \gcd(a, b) = d \quad \rightarrow \quad c|a \& c|b$$

$$\exists k \in \mathbb{Z} (ck = d)$$

$$\text{Also } d = \gcd(a, b) \text{ so } d|a \& d|b$$

$$\text{since } c|d \& d|a \text{ so } c|a$$

$$\text{since } c|d \& d|b \text{ so } c|b$$

$$\text{so if } c | \gcd(a, b) \iff c|a \& c|b \quad (\text{By Theorem 7.1.6, both sides hold})$$

These Theorems can be used to prove other theorems too.

7.1) 5)

$\rightarrow$ side follows as n is linear combination of a & b

$$\leftarrow d | n \quad \exists s \in \mathbb{Z} \exists t \in \mathbb{Z} (n = sa + tb)$$

$$\exists k \in \mathbb{Z} (dk = n) \text{ but } d = \gcd(a, b) \text{ so } \exists s, t \in \mathbb{Z} (d = sa + tb)$$

$$\text{so } n = dk = d(sa + tb) = ds a + dt b \quad \therefore n \text{ is a linear}$$

Combination of a & b .

7.1) 6) Let $d = \gcd(a, b)$. Then $d|a$ & $d|b$. Since $a+bc$ is linear combination of a & b , so $d|a+bc$.

Now supp $e|a+bc$ & $e|b \longrightarrow e \leq d$.

$$\exists k \in \mathbb{Z} (ek = a+bc)$$

$$\exists l \in \mathbb{Z} (el = b)$$

$$\text{so } ek = a+bc = a+elc$$

$$\text{so } ek - elc = a$$

$$\text{so } e(k - lc) = a \quad \text{so } e|a.$$

Also as $e|b$ so $e|\gcd(a, b) = d$ so $e \leq d$.

By Thm 7.1.6.

$$7.1) 8) \gcd(5a+2, 13a+5)$$

$$= \gcd(13a+5, 5a+2)$$

$$= \gcd(13a+5 - 2(5a+2), 5a+2) \quad \because \text{By Ex 6}$$

$$= \gcd(3a+1, 5a+2)$$

$$= \gcd(5a+2, 3a+1)$$

$$= \gcd(5a+2 - 1(3a+1), 3a+1) \quad \because \text{By Ex 6}$$

$$= \gcd(2a+1, 3a+1)$$

$$= \gcd(3a+1, 2a+1)$$

$$= \gcd(3a+1 - 1(2a+1), 2a+1) \quad \because \text{By ex 6}$$

$$= \gcd(a, 2a+1)$$

$$= \gcd(2a+1, a)$$

$$= \gcd(2a+1 - 2a, a) \quad \because \text{By ex 6}$$

$$= \gcd(1, a) = 1$$

7.1) 10) Let $d = \gcd(a, b)$ & $e = \gcd(na, nb)$

We will prove that $nd \leq e$ & $e \leq nd$ which proves $nd = e$.

We know $d|a$ & $d|b$

so $\exists j \in \mathbb{Z} (dj = a)$ & $\exists k \in \mathbb{Z} (dk = b)$

so $dnj = na$ & $dnk = nb$

so $dn|na$ & $dn|nb$. But e is the gcd, so $dn \leq e$.

We used the fact that any common divisor $\leq \gcd$.

Clearly $n|an$ & $n|bn$ so $n|\gcd(na, nb) = e$

so $\exists i \in \mathbb{Z} (ni = e)$ — ①

Also $e|na$ & $e|nb$ so $\exists l_1 \in \mathbb{Z} (el_1 = na)$ — ②

& $\exists l_2 \in \mathbb{Z} (el_2 = nb)$ — ③

① & ② $\rightarrow nil_1 = na \rightarrow il_1 = a$ so $i|a$

① & ③ $\rightarrow nil_2 = nb \rightarrow il_2 = b$ so $i|b$

But d is gcd of a & b so $i \leq d$. Multiplying by n ,
 $ni = e \leq dn$. Hence Proved.

Note: Formally

$d = \gcd(a, b) \rightarrow \forall c \in \mathbb{Z}^+ [(c|a \wedge c|b) \rightarrow c \leq d]$

7.1) 11) $\rightarrow \text{supp } d \in D[\gcd(a, b)]$. Then $d|\gcd(a, b)$

But $\gcd(a, b)|a$ & $\gcd(a, b)|b$ so $d|a$ & $d|b$ so
 $d \in D(a) \cap D(b)$.

$\leftarrow \text{supp } d \in D(a) \cap D(b)$ so $d|a$ & $d|b$. We already
know $d|\gcd(a, b)$ so $d \in D[\gcd(a, b)]$.

7.1) 11) let $d = \gcd(a, b)$ & $e = \gcd(\gcd(a, b), c)$

$$d|a \text{ \& } d|b \text{ \& } e|d = \gcd(a, b) \text{ \& } e|c$$

$$\text{so } e|a \text{ \& } e|b \text{ \& } \text{so } e \in D(a) \cap D(b) \cap D(c).$$

let j be arbitrary & supp $j \in D(a) \cap D(b) \cap D(c)$

$$\text{Then } j|a \text{ \& } j|b \text{ \& } j|c$$

$$\text{so } j|\gcd(a, b) = d$$

$$\text{so } j|d. \text{ Also } j|c \text{ but } c \text{ is } \gcd(d, c) \text{ so } j \leq c.$$

so c is the largest element of $D(a) \cap D(b) \cap D(c)$

7.1) 12)(a) $55 = 34(1) + 21 \rightarrow F_1$

$$34 = 21(1) + 13$$

$$F_8 = 34$$

$$21 = 13(1) + 8$$

$$F_9 = 55$$

$$13 = 8(1) + 5$$

8 Division Steps.

$$8 = 5(1) + 3$$

$$5 = 3(1) + 2$$

$$3 = 2(1) + 1 \rightarrow F_1 = \gcd$$

$$2 = 1(2) + 0 \rightarrow F_0$$

$$F_{n+1} = F_n + F_{n-1}$$

$$q \neq 1 \text{ as } 2F_n > F_{n+1}$$

$$\text{as } F_{n-1} < F_n$$

(b) $\gcd(F_{n+1}, F_n) = 1$ & there are n divisions

7.1) 14) let $b \in \mathbb{Z}^+$ be arbitrary. we will use strong induction on a . supp

$$\forall k \in \mathbb{Z}^+ [k \in \mathbb{Z}^+ \rightarrow \exists q \in \mathbb{N} \exists r \in \mathbb{N} (k \leq \frac{b}{2} \wedge (k = qb + r \vee k = qb - r))]$$

we will prove this for a .

①

case 1) $a - b > 0$ or $a > b$

$$\text{By ①} \rightarrow (a - b = qb + r \vee a - b = qb - r) \wedge r \leq \frac{b}{2}$$

$$\text{case 1.1) } a - b = qb + r$$

$$\text{then } a = (q+1)b + r \text{ \& } q+1 \in \mathbb{N} \text{ clearly}$$

$$\text{case 1.2) } a - b = qb - r$$

then $a = (q+1)b - r$ & $q+1 \in \mathbb{N}$.

Case 2) $a \leq b$

In this case, $a-b \notin \mathbb{Z}^+$ so we cannot apply inductive hypothesis. But we can compute some examples & note

supp $a=14$ & $b=28$

$$\text{then } 14 = 28(0) + 14 \quad \& \quad 14 \leq r = a \leq \frac{b}{2} \quad \text{--- ①}$$

if $b=26$, then $14 = 26(0) + 14$ is not possible as $r = 14 > \frac{b}{2}$

$$\text{so } 14 = 26(1) - 12 \quad \& \quad r = 12 = b - a \leq \frac{b}{2} \quad \text{--- ②}$$

Note in ①, $a \leq \frac{b}{2}$ & in ② $a > \frac{b}{2}$. This motivates 2 more cases.

Case 2.1) $a \leq \frac{b}{2}$

$$\text{then } a = 0b + a \quad \& \quad r = a \leq \frac{b}{2} \quad \& \quad 0 \in \mathbb{N}$$

Case 2.2) $a > \frac{b}{2}$

$$\text{then } -a < -\frac{b}{2} \rightarrow b-a < b - \frac{b}{2} = \frac{b}{2}$$

so $a = 1b - (b-a)$ where $1 \in \mathbb{N}$ clearly & as shown,

$$r = b-a < \frac{b}{2}$$

7.1) 14)(b) Supp $d = \gcd(a, b)$, then

$$d|a \rightarrow \exists j \in \mathbb{Z} (dj = a) \quad \text{--- ①}$$

$$d|b \rightarrow \exists k \in \mathbb{Z} (dk = b) \quad \text{--- ②}$$

We will first show that $d|r$

Case 1) $a = qb + r$

$$dj = qdk + r$$

$$r = dj - qdk$$

$$r = d(j - qk) \quad \& \quad j - qk \in \mathbb{Z} \quad \text{so } d|r$$

Case 2) similar & yields $r = d(qk - j)$

Now supp $c|b$ & $c|r$, where c is arbitrary $\rightarrow c \leq d$

Another way to understand this case is to plot $\frac{b}{2}$ on a number line

then $\exists m \in \mathbb{Z} (cm = b)$

$\exists n \in \mathbb{Z} (cn = a)$

Case 1) $a = qb + r$

$$a = q(cm) + cn$$

so $a = c(qm+n)$ so $c|a$. As $c|b$ too but $d = \gcd(a,b)$

so $c \leq d$.

Case 2) similar $\rightarrow$ yields $a = c(qm-n)$ so $c|a$.

7.1) 14) c) similar to Euclidean Algorithm. If $\neg \exists m (r_m = 0)$

then as for every n , $r_{n+1} \leq \frac{r_n}{2}$, the list $r_0, r_1, r_2, \dots$ would be a decreasing list of natural numbers with no least element which contradicts the well ordering principle.

so $\exists m (r_m = 0)$. Applying Ex 14(a) to each pair of remainders, we get:

$$\gcd(a,b) = \gcd(r_0, r_1) = \gcd(r_1, r_2) = \dots = \gcd(r_{m-2}, r_{m-1}) = r_{m-1}$$

7.1) 14) d) Euclidean Algorithm

Least Remainder Absolute

$$1515 = 555(2) + 405$$

$$1515 = 555(3) - 150$$

$$555 = 405(1) + 150$$

$$555 = 150(4) - 45$$

$$405 = 150(2) + 105$$

$$150 = 45(3) + 15$$

$$150 = 105(1) + 45$$

$$45 = 15(3) + 0$$

$$105 = 45(2) + 15$$

$$45 = 15(3) + 0$$

$$\gcd(1515, 555) = 15$$

Least Remainder takes fewer steps because each remainder is strictly $\leq$ half of previous. Therefore reaching 0 quickly.